

Investigating if Resveratrol Can Reduce Seizure-like Activity in Shaker Mutant Drosophila by Reducing Oxidative Stress

Background:

Oxidative Stress & Inflammation

Epilepsy and seizure disorders are linked to **neural hyperexcitability**, often modeled using **Drosophila melanogaster Shaker mutants** that have defective voltage-gated potassium channels. These mutants exhibit a low threshold for mechanically induced seizures, a phenotype exacerbated through **oxidative stress**. This study investigates whether the antioxidant polyphenol **Resveratrol** provides neuroprotection and reduces seizure severity by **neutralizing oxidative damage** induced by **hydrogen peroxide**.

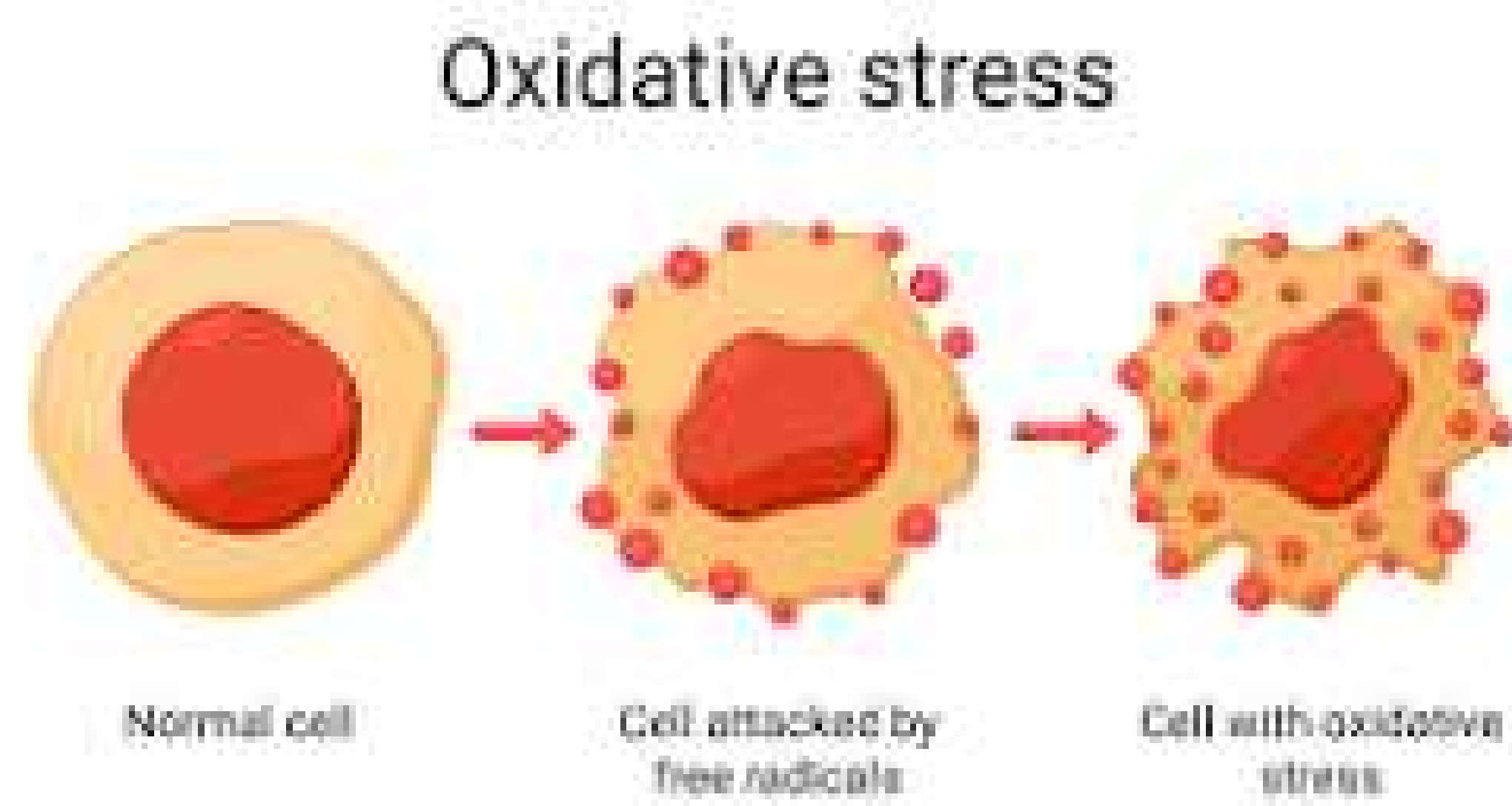


Figure 1: Oxidative stress effect on cells

Shaker mutant

The shaker mutant of *Drosophila melanogaster* has a genetic defect in a **voltage-gated potassium ion channel**, leaving neurons highly susceptible to **hyperexcitability** and **mechanically induced seizures**.

Resveratrol

Resveratrol is a natural polyphenolic antioxidant known to neutralize reactive oxygen species and **mitigate oxidative stress**.

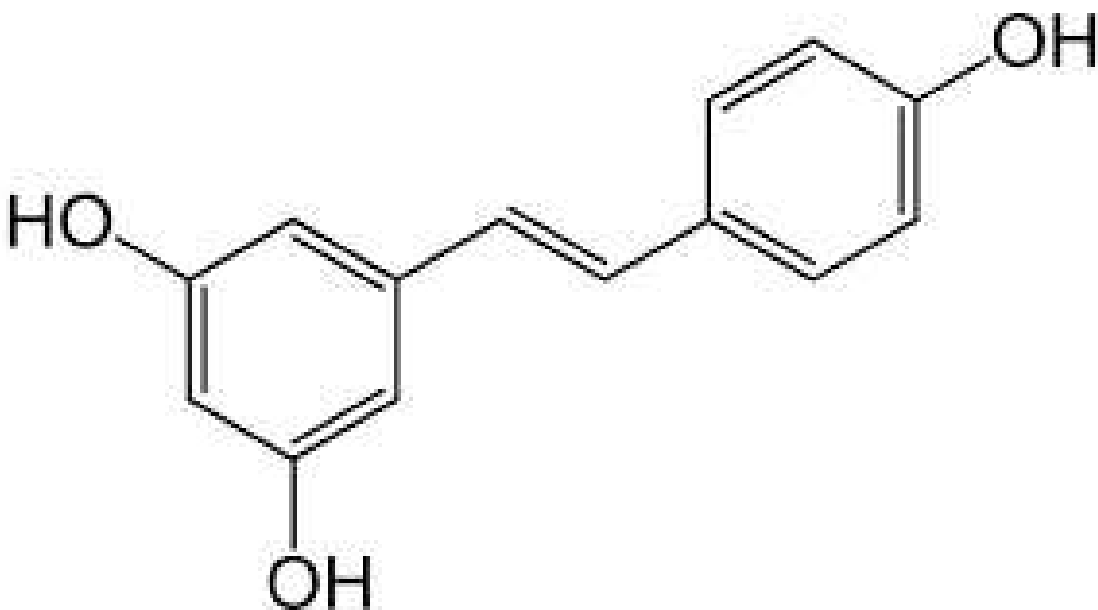


Figure 2: Resveratrol chemical structure

- The study will investigate **Resveratrol's** potential to act as an antioxidant agent via buffering neurons against oxidative damage caused by hydrogen peroxide
- Since **Potassium channel mutations** destabilize membrane potentials, using dietary antioxidants to minimize environmental stressors offers a therapeutic strategy to lower seizure recovery times and keep motor stability

Hypothesis

Purpose: The objective of this research is to evaluate the therapeutic potential of resveratrol in reducing the duration and frequency of seizures in Shaker mutant flies. By using hydrogen peroxide as a pro-oxidant, this experiment aims to isolate whether Resveratrol's primary benefit is derived from its ability to manage oxidative stress.

Hypothesis: If Shaker mutant *Drosophila melanogaster* are exposed to oxidative stress via dietary Hydrogen Peroxide (30%), then their mechanically induced seizure recovery duration will significantly increase due to heightened neural hyperexcitability; however, pre-treatment with 100 μ M Resveratrol will rescue this phenotype and minimize recovery time by neutralizing reactive oxygen species.

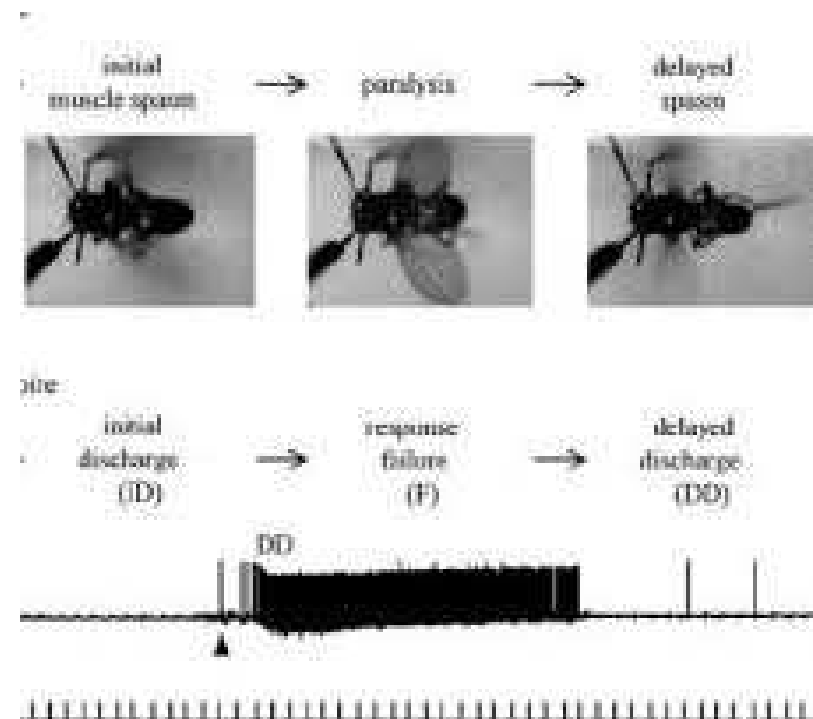


Figure 3: Fly groups

Methods

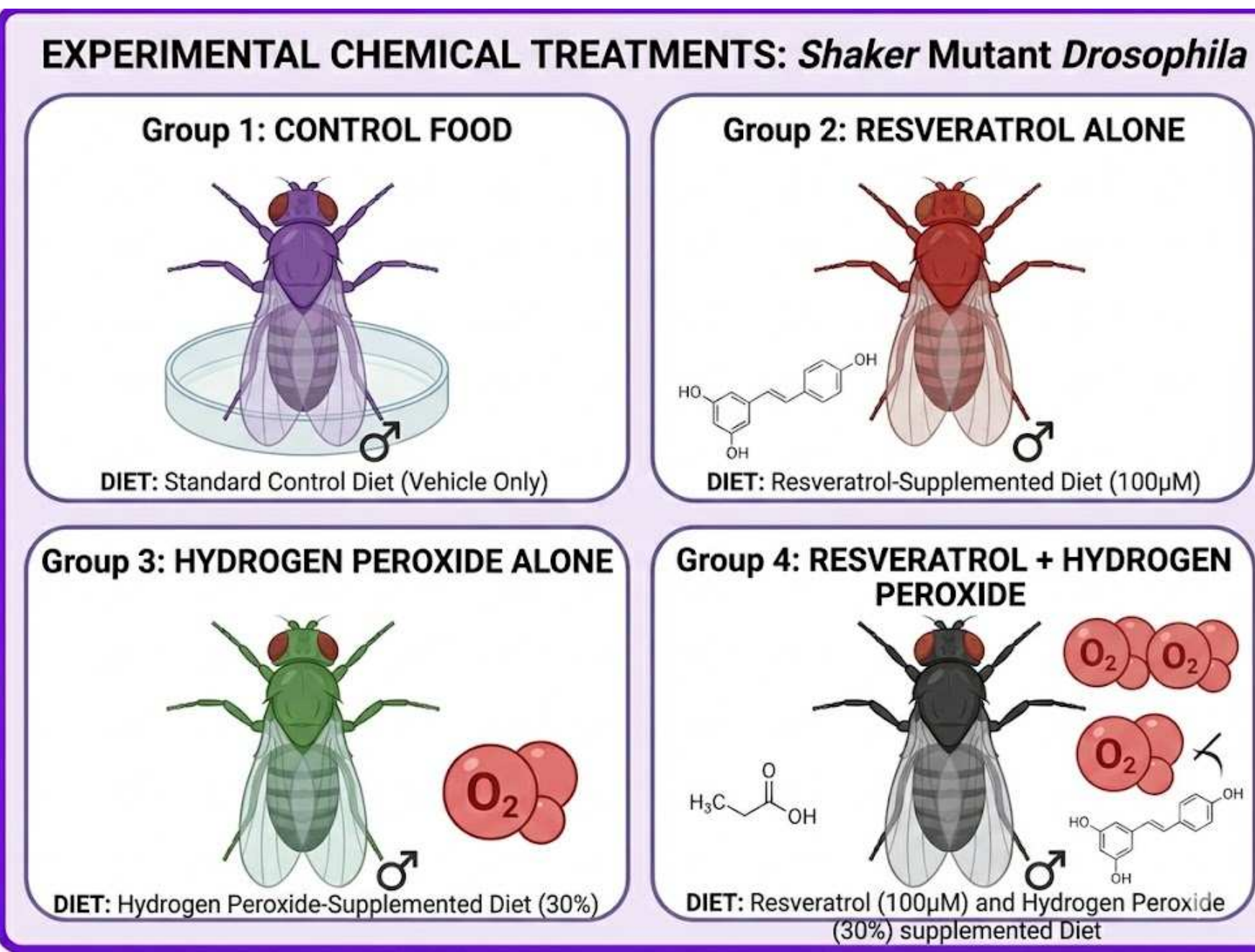


Figure 5: Experimental Groups

4 Groups of *Drosophila* will be given different combinations of chemicals and exposed to them for 48 hours

Independent variables: Drug given to flies

Dependent variable: Seizure duration

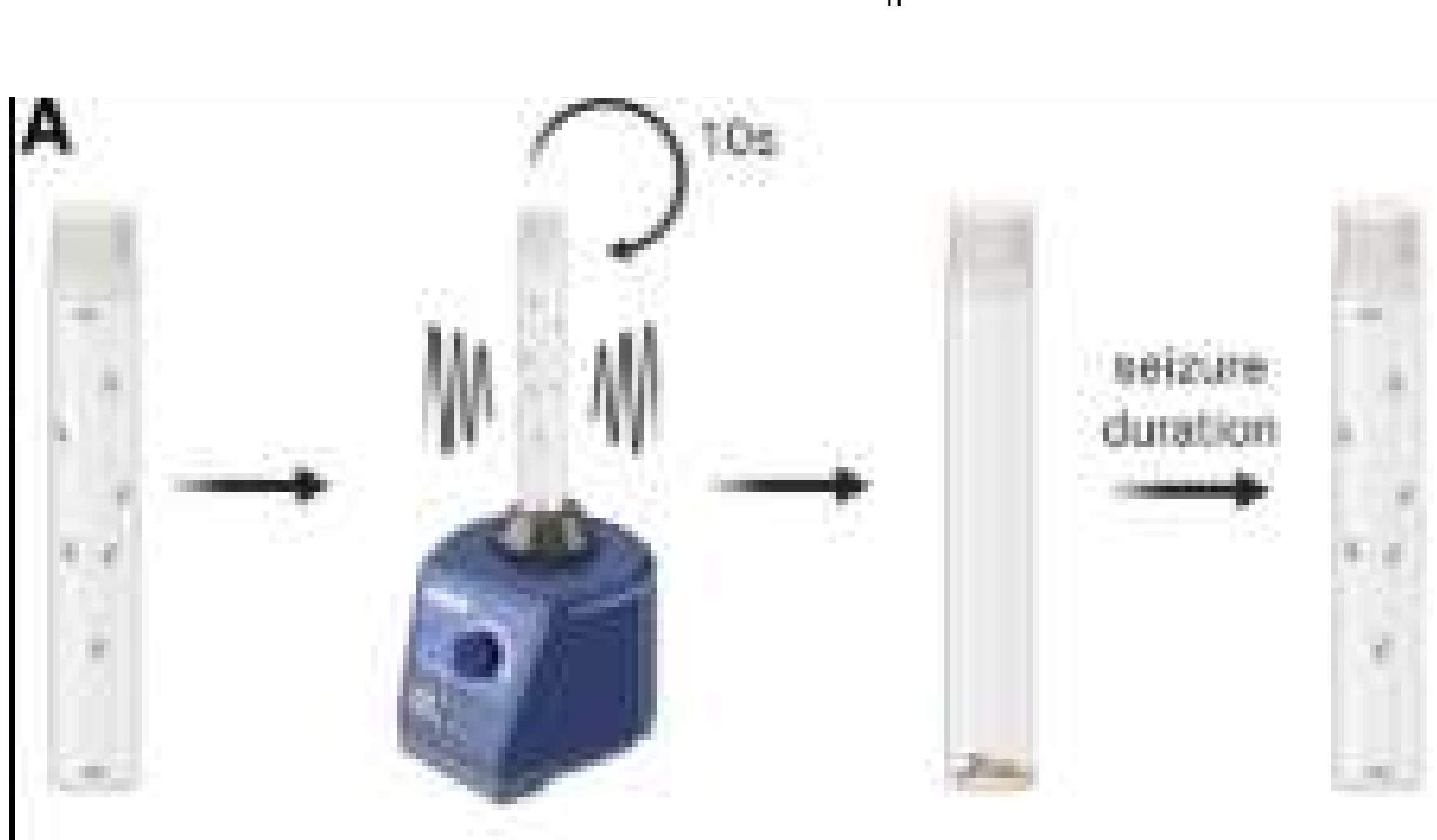


Figure 5: Adult vortex assay

Adult Vortex assay will be used to induce a mechanical shock into the *Drosophila*

Results

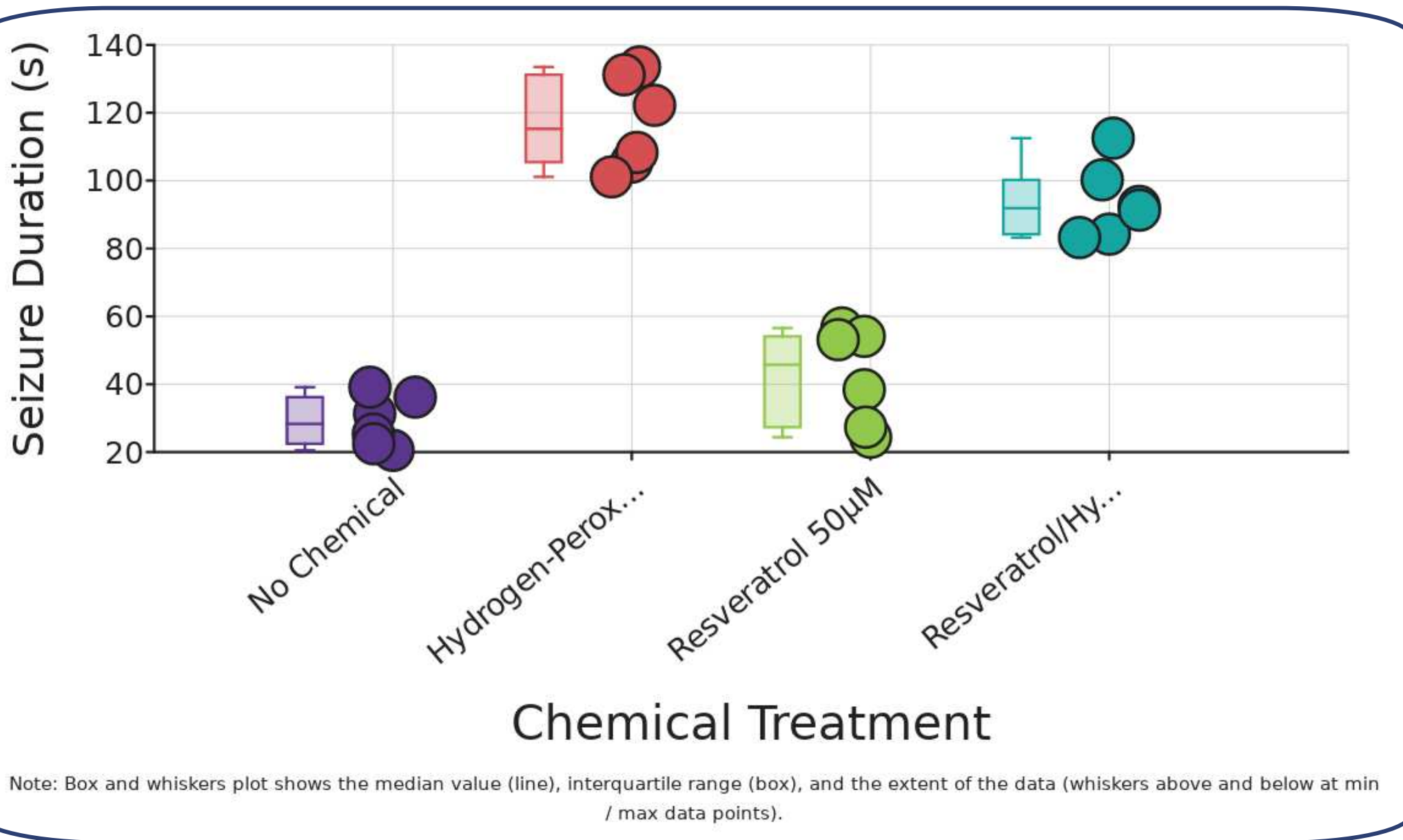


Figure 7: Graph

Effects of Hydrogen Peroxide and Resveratrol on Mechanical Seizure Duration in Shaker Mutant Drosophila				
		50 μ M	100 μ M	
	No Chemical	Hydrogen-Peroxide	Resveratrol	Resveratrol/Hydrogen-
Trial 1	20.51	105.47	24.38	84.23
Trial 2	31.32	133.51	38.37	92.42
Trial 3	36.2	122.21	56.56	83.23
Trial 4	25.32	131.22	27.34	91.32
Trial 5	22.45	108.32	54.11	112.52
Trial 6	39.12	101.12	53.12	100.24
Mean	29.15	116.98	42.31	93.99
Standard	6.93	12.66	13.04	10.98

Figure 6: Data table

Conclusions:

Statistical justification: A parametric one-way analysis of variance (ANOVA) was used to analyze the experiment, as it evaluates variance across multiple groups simultaneously, and helps in concluding if the recovery differences between the groups are statistically significant or if aren't.

Effect	Degrees of Freedom (df)	F-statistic (MS / MS residual)	P-value	Interpretation of P
Chemical Treatment	3	73	<0.01	A P-value of <0.01 means extremely strong confidence that the groups are different.
Error or Residual	20			

Figure 8: Anova Test

Since the ANOVA test determined a significant difference between the recovery time within the groups, **A Post-hoc Tukey Range Test** was run to determine which groups were significantly different from each other.

Pair	Mean diff	P-value	Interpretation of P
Hydrogen-Peroxide 50 μ M (30%) - No Chemical	87.8	<0.01	Means: extremely strong confidence that the groups are different
Resveratrol 50 μ M - No Chemical	13.2	0.26	Means: little to no indication that the groups might be different
Resveratrol/Hydrogen-Peroxide 100 μ M (30%) - No Chemical	64.8	<0.01	Means: extremely strong confidence that the groups are different
Resveratrol 50 μ M - Hydrogen-Peroxide 50 μ M (30%)	-74.7	<0.01	Means: extremely strong confidence that the groups are different
Resveratrol/Hydrogen-Peroxide 100 μ M (30%) - Hydrogen-Peroxide 50 μ M (30%)	-23	0.02	Means: very strong confidence that the groups are different
Resveratrol/Hydrogen-Peroxide 100 μ M (30%) - Resveratrol 50 μ M	51.7	<0.01	Means: extremely strong confidence that the groups are different

Figure 9: Post Hoc Tukey Range test

Conclusion: While there was a significant difference between the No chemical group and the Resveratrol to the Hydrogen peroxide group, and the Hydrogen Peroxide-Resveratrol group, there was no significant difference between these subsets of groups, as seen from Figure 7. However, the means of the Resveratrol group were less than those of the hydrogen peroxide group, as seen from Figure 6. This supports that while Hydrogen peroxide significantly increased seizure frequency, Resveratrol was not shown to reduce oxidative stress from the control group or the hydrogen peroxide group. This may have resulted from limitations in the experiment, such as tracking software errors which restricted testing to a single fly at a time, limiting the amount of data able to be collected, alongside time. Constraints resulted from errors in procedures.

Citations

- Smoot J, Popodi S, Colbourn E, et al. Assessment of chemical toxicity in adult *Drosophila melanogaster*. J Vis Exp. 2023;(193):e65029. doi:10.3791/65029
- Aggarwal A, Reichert H, VijayaRaghavan K. A locomotor assay reveals deficits in heterozygous Parkinson's disease model and proprioceptive mutants in adult *Drosophila*. Proc Natl Acad Sci U S A. 2019;116(49). doi:10.1073/pnas.1807456116
- Scheunemann L, Skrobilin T, Simpson C, Tanimoto H. The propensity for consuming ethanol in *Drosophila* requires rutabaga adenylyl cyclase expression within mushroom body neurons. Front Behav Neurosci. 2018;12:223. doi:10.3389/fnbeh.2018.00223
- Liu Q, Tian J, Yang X, Li Y, Aike G. Locomotor assay in *Drosophila melanogaster*. Bio-Protoc. 2017;7(10):e2283. doi:10.21769/BioProtoc.2283
- Kruger L, Denton T. A standardized method for incorporation of compounds into food for use with *Drosophila melanogaster*. Anal Biochem. 2020;599:113740. doi:10.1016/j.ab.2020.113740
- Mtuzaithe J, Petersen R, Chang A, Baines R. Characterization of seizure induction methods in *Drosophila*. eNeuro. 2021;8(3). doi:10.1523/ENEURO.0079-21.2021